

Ryan Tang

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Education

Massachusetts Institute of Technology

B.S. Electrical Engineering with Computing and Physics, GPA 4.3/5.0

Cambridge, MA

Expected May 2029

- **Coursework:** Circuits & Electronics, C & Assembly, Computation Structures*, Signal Processing*, Algorithms*, Probability*, Waves & Vibrations, Linear Algebra, Differential Equations *in progress

Experience

MIT Satellite Team

Jan. 2026 – Present

Instrumentation Lead

Cambridge, MA

- Leading design of a UV-visible (200–500 nm) spectrometer payload, covering the optics, enclosure, electronics, firmware, and desktop software. An early prototype flew on a high-altitude balloon to 100,000 ft
- Optimized the Czerny-Turner layout in Zemax to get a focused spot under 20 μm , smaller than a detector pixel
- Designed three revisions of a compact 3D-printed enclosure with kinematic mirror and detector mounts, a slit mount, and baffles to block stray light
- Found and fixed signal integrity problems on the first board. Redesigned the analog front end and low-noise supplies, and split analog and digital onto separate boards. Raw output now sits at the ADC noise floor without averaging
- Moved sensor timing into a CPLD and wrote the C++ firmware, with zero dropped pixels over 900+ frames
- Wrote a Python desktop app for live spectra, wavelength calibration, and data export
- Currently laying out a 4-layer third revision with the microcontroller on board

MIT Quantum & Precision Measurements Group (Prof. Vivishek Sudhir)

Jan. 2026 – Present

Undergraduate Researcher

Cambridge, MA

- Built a 1550 nm Michelson interferometer with a piezo-driven mirror, read out through a Red Pitaya FPGA board. It is a prototype for a 9-interferometer sensor array for active seismic isolation
- Wrote Python processing to calibrate the interferometer and model the piezo's nonlinearity and hysteresis. The model predicted measured motion with $R^2 = 0.997$
- Wrote PID control to lock the interferometer at quadrature and hold the mirror at set positions to within 1.7 nm RMS
- Measured the system's displacement noise floor and compared it to standard seismic noise models

Projects

RISC-V CPU | Verilog, Vivado, Spartan-7 FPGA

Aug. – Sep. 2026

- Wrote a 5-stage pipelined RV32I processor with hazard handling, traps, and memory-mapped I/O, plus 24 testbenches

FPGA Neural Network Accelerator | Verilog, Python, Spartan-7 FPGA

Sep. 2026

- Built an 8×8 systolic array that classifies handwritten digits sent over UART from a desktop app, at 94% accuracy

Datasheet Reviewer | Python, Claude, PyMuPDF, TypeScript

Aug. 2026

- Built an AI tool that reads component datasheets and saves pinouts, electrical specs, timing, and layout guidance to a parts library. It answers schematic design questions across the library and cites the datasheet page

Homemade Semiconductors and 8-Bit Computer | Process development, 74LS logic

2023 – 2024

- Made working PN diodes in a garage (thermal oxidation, HF etch, diffusion doping) and built an 8-bit TTL computer

Skills & Activities

Electrical: KiCad, mixed-signal PCB layout, low-noise analog design, power supplies, board bring-up and debugging

Embedded: C/C++, RP2040/RP2350 (PIO), SPI, I²C, UART, CPLDs, Verilog, Vivado

Optics: Zemax OpticStudio, spectrometer design, tolerancing, stray light control, interferometry

Mechanical: Onshape, optomechanical and kinematic design, flexures, stress analysis, material selection, 3D printing

Software: Python (NumPy, SciPy), PID control, Git, Linux

Activities: Director of MakeMIT (\$60k budget), MIT Motorsports drivetrain, 2 $\times$ USNCO Finalist, 2 $\times$ CERN Beamline for Schools shortlist